

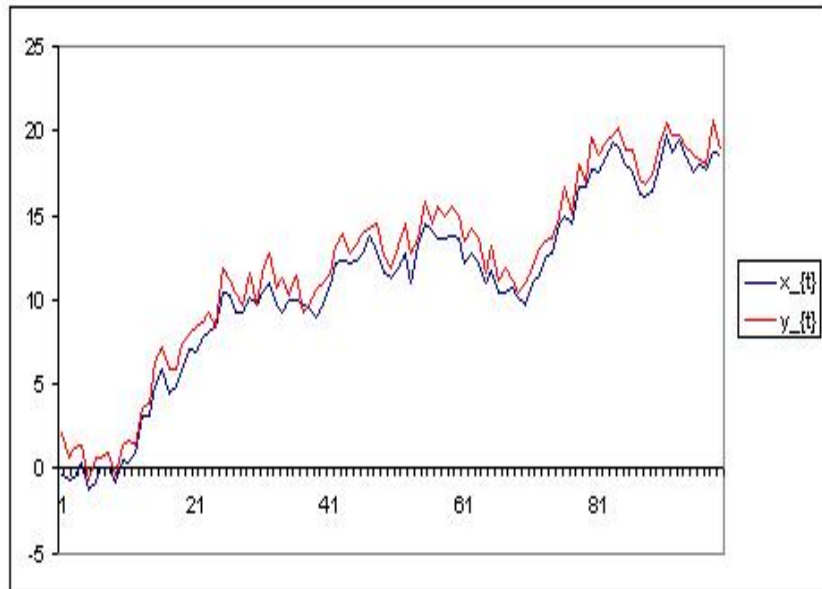
Advanced Econometrics Lecture 13

dr Marcin Chlebus, dr Rafał Woźniak

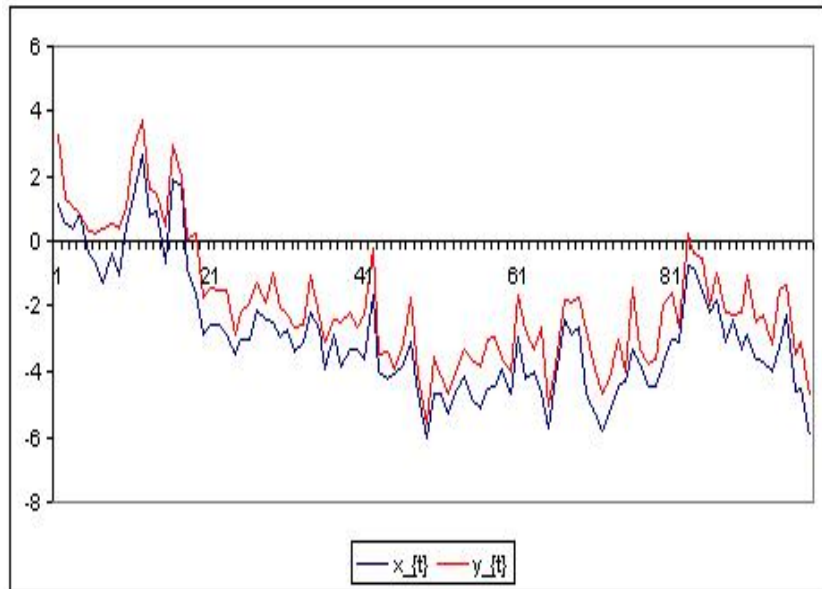
Faculty of Economic Sciences, University of Warsaw

Warsaw, 26-05-2020

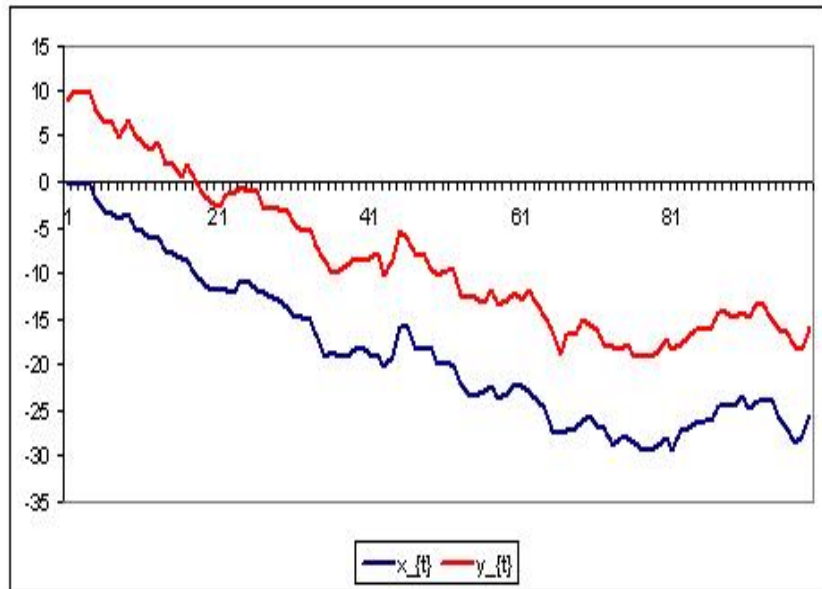
Exercise 1. Are the series cointegrated?



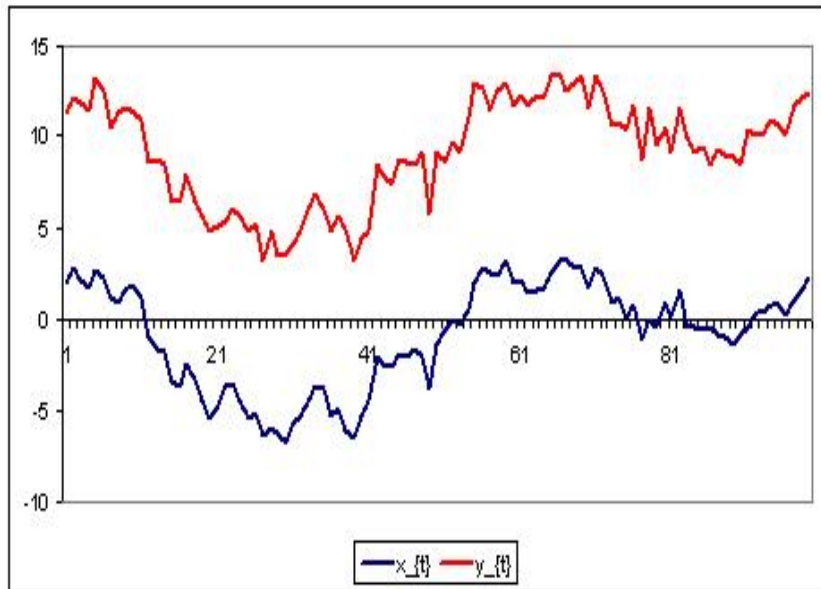
Exercise 2. Are the series cointegrated?



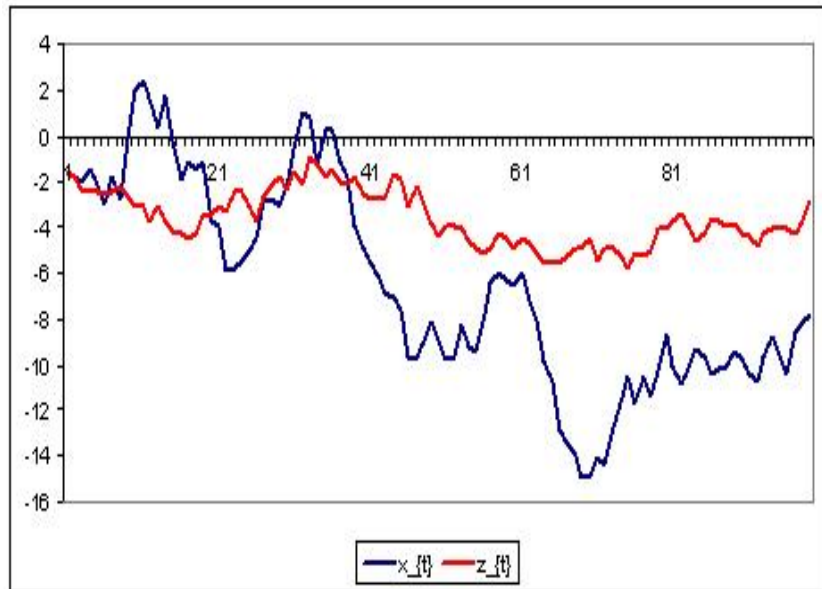
Exercise 3. Are the series cointegrated?



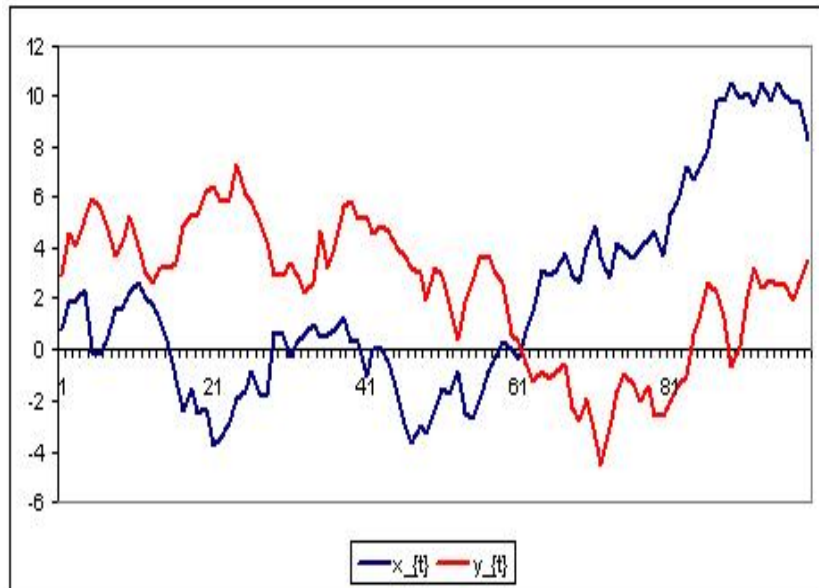
Exercise 4. Are the series cointegrated?



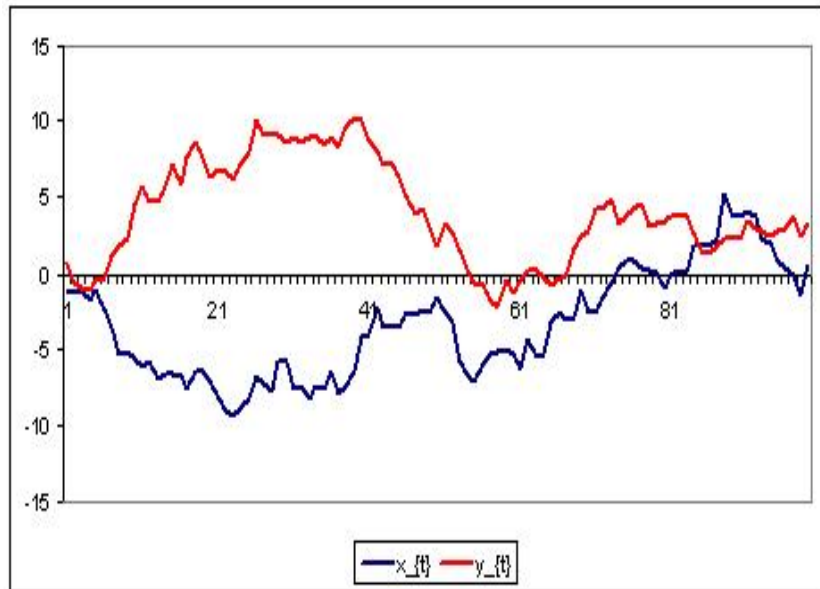
Exercise 5. Are the series cointegrated?



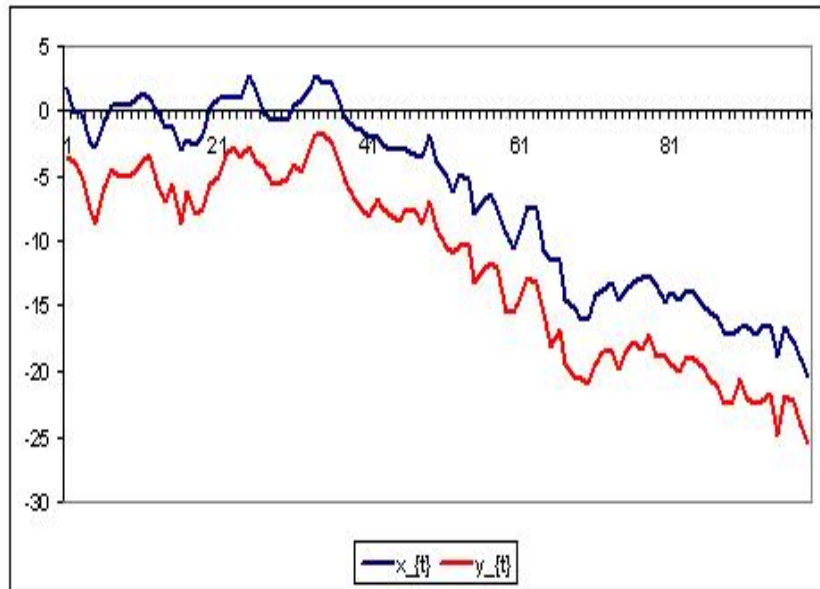
Exercise 6. Are the series cointegrated?



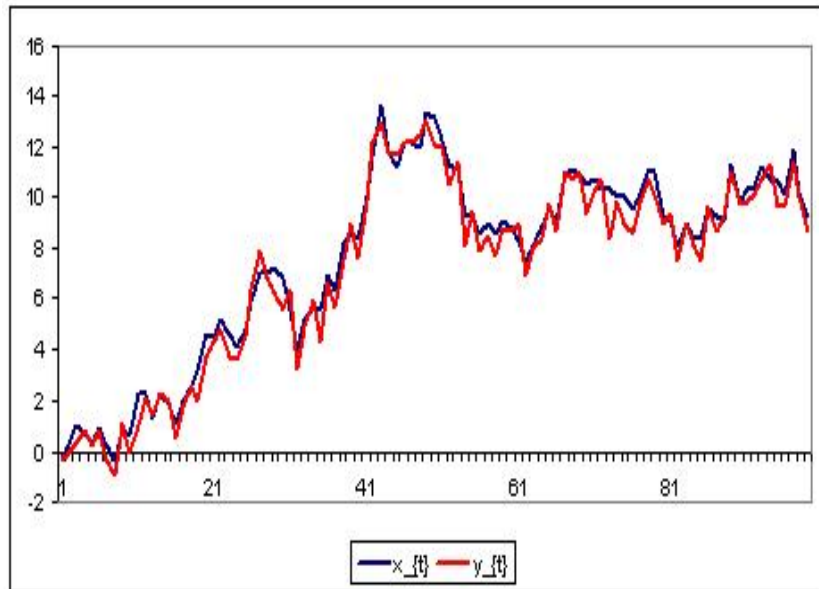
Exercise 7. Are the series cointegrated?



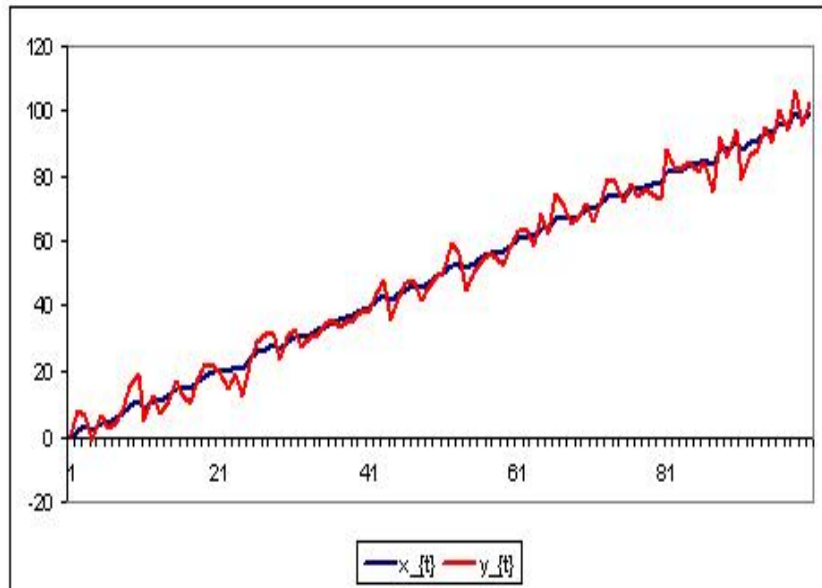
Exercise 8. Are the series cointegrated?



Exercise 9. Are the series cointegrated?



Exercise 10. Are the series cointegrated?



Are the series cointegrated?

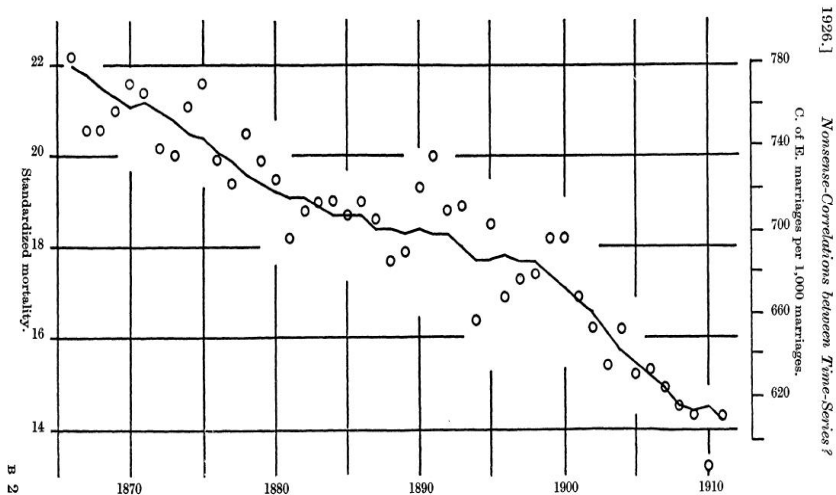
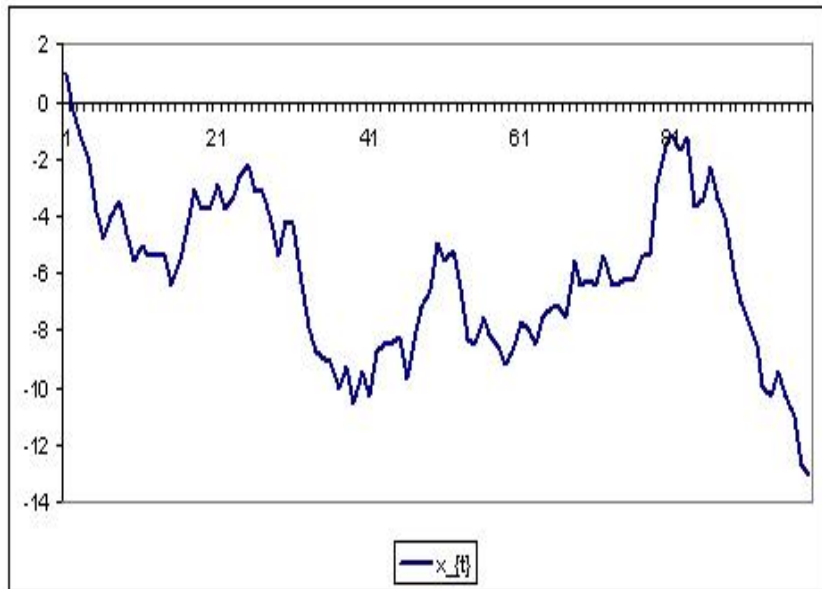


FIG. 1.—Correlation between standardized mortality per 1,000 persons in England and Wales (circles), and the proportion of Church of England marriages per 1,000 of all marriages (line), 1866–1911. $r = +0.9512$.

- Correlation between standardized mortality per 1,000 persons in England and Wales (circles), and the proportion of Church of England marriages per 1,000 of all marriages (line), 1866-1911.
- $r = +0.9512$ (Correlation)
- Źródło: G. Udny Yule, **Why do we Sometimes get Nonsense-Correlation between Time-Series?—A Study in Sampling and the Nature of Time-Series**, Journal of the Royal Statistical Society, Vol. 89, No. 1 (Jan., 1926), pp.1-63.

- Another example: Price Level and Cumulative Rainfall in UK.
- David F. Hendry, **Econometrics-Alchemy or Science?**, *Economica*, New Series, Vol. 47, No. 188 (Nov., 1980), pp. 387-406.
- This relation proved resilient to many econometric diagnostic tests and was humorously advanced as a new theory of inflation.

Random Walk - A Drunk ($y_t = y_{t-1} + \epsilon_t$)



- Michael P. Murray, **A Drunk and Her Dog: An Illustration of Cointegration and Error Correction**, The American Statistician, February 1994, Vol. 48, No. 1.
- Aaron Smith and Robin Harrison, **A Drunk, Her Dog and a Boyfriend: An Illustration of Multiple Cointegration and Error Correction**.

The Tale of the Drunk and Her Dog

- ... the dog belongs to the drunk.
- The drunk sets out from the bar, about to wander aimlessly in random-walk fashion.
- But periodically she intones "Oliver, where are you?", and Oliver interrupts his aimless wandering to bark.
- He hears her; she hears him.
- He thinks, "Oh, I can't let her go too far off; she'll lock me out."
- She thinks "Oh, I can't let him get too far off; he'll wake me up in the middle of the night with his barking."
- Each assesses how far away the other is and moves to partially close the gap.

The Tale of the Drunk and Her Dog

- Now neither drunk nor dog follows a random walk; each has added what we formally call an *error-correction mechanism* to her or to his steps.
- Significantly, despite the nonstationarity of the paths, one might still say, "If you find her, the dog is unlikely to be very far away."
- If this is right, then the distance between the two paths is stationary, and the walks of the woman and her dog are said to be *cointegrated of order zero*.

Brexit and the markets – The Economist (link)

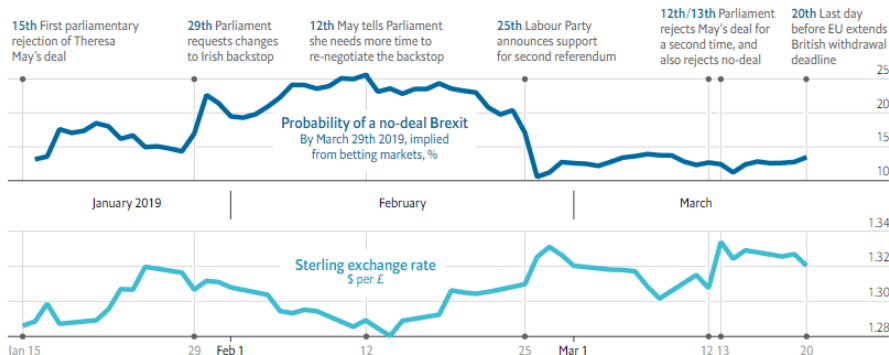
REGARDLESS OF WHAT they tell you, traders struggle to explain short-term fluctuations in the value of currencies. Recently, however, the pound has become an exception. Every time it seems more likely that Britain will leave the EU without a deal, sterling falls against the dollar.

The strength of this link can be measured statistically, thanks to a helpful proxy for the odds of no-deal. On January 16th a market opened on Betfair Exchange, a betting website, on whether Britain will crash out by March 29th, the original Brexit deadline. Punters have bet £3.9m (\$5.1m).

On March 21st the EU extended this deadline, causing the chances of no-deal by the end of March to fall to near zero. But for the 64 days between the opening of the market and the granting of the extension, the odds seemed to mirror the exchange rate. For each ten-percentage-point rise in the probability of no-deal, the pound lost \$0.02, and vice versa. As sterling moved between \$1.28 and \$1.33, it was possible to predict the exchange rate from Betfair's odds with an average error of just one cent.

Brexit and the markets – The Economist (link)

Fear of a no-deal Brexit has been driving the price of sterling



Britain

Aug 13th 2016 edition >

Thinner pets

Subwoofers

The curious incident of the dog and the waistline

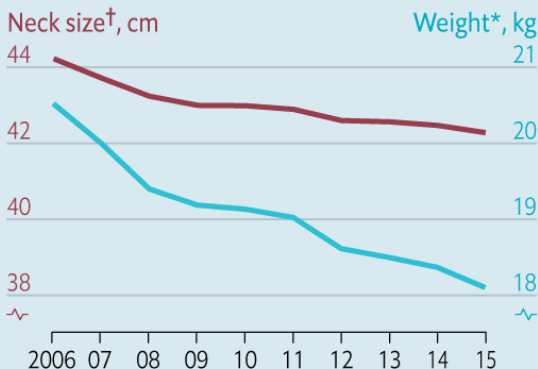
WHILE their owners bulge, British dogs appear to be shrinking. Each year the Kennel Club, a dog-lovers' association, registers the details of about 250,000 hounds; the club believes it has records for about one-third of the dogs in Britain. An analysis by *The Economist* using data supplied by the club suggests that the weight of the average British pup has fallen by about 12% in the past decade (see chart). As smaller breeds grow in popularity, the average size of a dog, as measured by the circumference of its neck, has also fallen.



Thinner pets – The Economist (link)

Fit as a butcher's dog

Characteristics of dogs registered with the UK's Kennel Club, average when fully grown

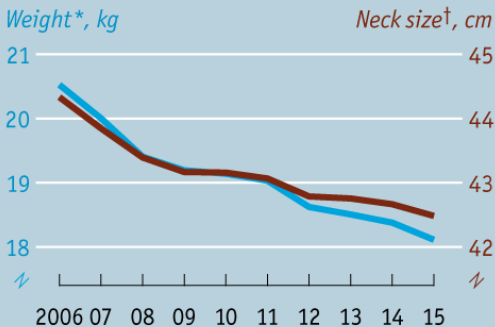


Sources: Kennel Club; *The Economist* *Where at least 50 are registered per year
†Where at least 100 are registered per year

Thinner pets – The Economist (link)

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Business

Feb 29th 2020 edition >

Industrial light and magic

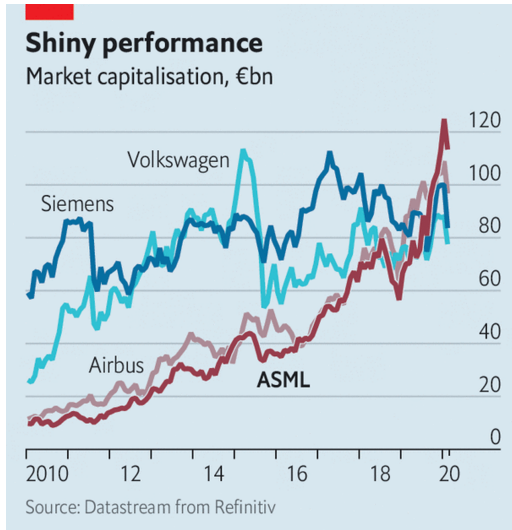
How ASML became chipmaking's biggest monopoly

A low-key Dutch company has cornered a critical link in the global electronics supply chain

Chip making industry – The Economist (link)

Ask people to pinpoint the centre of the digital economy and many will finger Silicon Valley, populated by Apple, Google, Facebook and too many sexy startups to count. Others may nod at the area around Seattle, where Amazon and Microsoft are based. Some could suggest Shenzhen, China's technology hub. Few would point to a nondescript suburb of Eindhoven, the Netherlands' fifth-biggest city. Yet on closer inspection, the case for Veldhoven looks compelling. It is home to asml, the world's sole manufacturer of the most advanced equipment critical to modern chipmaking. If chips make the world go round, asml may be the closest the multi-trillion-dollar global tech industry has to a linchpin.

Chip making industry – The Economist (link)



The Economist